

Technical Document #850: Blockchain - Comprehensive Overview

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Decentralized finance (DeFi) recreates traditional financial systems using blockchain technology. It enables lending, borrowing, and trading without intermediaries.

Cross-chain technology enables communication between different blockchains. It is essential for interoperability in the blockchain ecosystem.

Consensus mechanisms ensure all nodes in a network agree on the state of the blockchain. They are essential for maintaining network integrity.

Smart contracts are self-executing contracts with terms directly written into code. They automatically enforce agreements without intermediaries.

Sustainability and environmental responsibility are becoming key considerations in technology design. Green computing aims to reduce the environmental impact of technology.

Edge computing processes data closer to its source, reducing latency and bandwidth usage. It is essential for real-time applications like autonomous vehicles.

Federated learning trains models across decentralized data sources without sharing raw data. It enables privacy-preserving machine learning.

Key Takeaways:

- Cross-chain technology enables communication between different blockchains.
- Smart contracts are self-executing contracts with terms directly written into code.
- Consensus mechanisms ensure all nodes in a network agree on the state of the blockchain.